R 位编码，共 2^R 个结果 $\xrightarrow{X}\boxed{A/D}\xrightarrow{\hat{X}}$

Distorsion $D = E[(X-\hat{X})^2]$

[例] Uniform $X \sim U[-A, A]$

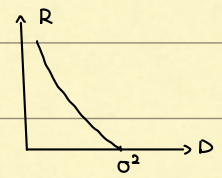
共 2^R 个格点，$\Delta = \frac{2A}{2^R}$，$D = \frac{1}{12}\Delta^2 \propto 2^{-2R}$

Rate Distorsion Theory

$$R = \max_{p(\hat{x}|x)} I(X;\hat{X}) \quad \text{s.t.} \quad E[(X-\hat{X})^2] \leq D$$

[例] Gaussian $X \sim N(0, \sigma^2)$

$$R = \begin{cases} \frac{1}{2}\log \sigma^2/D, & D \leq \sigma^2 \\ 0, & D > \sigma^2 \end{cases} \qquad D = \sigma^2 2^{-2R} \propto 2^{-2R}$$



Pf：当 $D > \sigma^2$ 时，直接 $\hat{X} = 0$ 即无需编码

当 $D \leq \sigma^2$ 时，

$$R = \max_{p(\hat{x}|x)} I(X;\hat{X}) = \max_{p(\hat{x}|x)} H(X) - H(X|\hat{X})$$

$$= \frac{1}{2}\log 2\pi e\sigma^2 - \max_{p(\hat{x}|x)} H(\hat{X}-X|\hat{X})$$

$$H(X-\hat{X}|\hat{X}) \overset{\text{indp}}{\leq} H(X-\hat{X}) \overset{\text{Gaussian}}{\leq} H(N(0, E[(X-\hat{X})^2]))$$

$$= \frac{1}{2}\log 2\pi e\sigma^2 - \frac{1}{2}\log 2\pi e D$$

$$= \frac{1}{2}\log \frac{\sigma^2}{D}$$

$$\begin{array}{ccccc} & & N(0,D) & & \\ & & \downarrow & & \\ \hat{X} & \longrightarrow & \oplus & \longrightarrow & X \\ \sim N(0, \sigma^2-D) & & & & \sim N(0, \sigma^2) \end{array}$$